

looking to cut down its electricity consumption and build a green office environment at its India branch. Shriyuj, a solutions provider formerly known as Varad Electro Systems, went ahead with a mix of motion/occupancy sensors, timer-based controls and a few more lighting automation solutions. The setup allowed electricity usage control based on holidays and work-hours. Initially installed to control lighting, the application was upgraded to include AC controls as well. As a result, the company achieved savings of ₹ 25,000 in the first month's electricity bill, which increased up to ₹ 47,000 in the fifth month. Bekaert reported up to 25 per cent electricity savings throughout the year, which cut down the costs by nearly ₹ 380,000 in one year. Additionally, reduced electricity consumption cut down carbon dioxide emission by 24 tonnes.

Intelligent usage only for the people. Motion sensors are used to switch on power only when a human is detected in the area covered. Technology has advanced enough to differentiate between humans, animals and other objects during detection. Moreover, controls can be customised to run lighting based on the availability of daylight. Organisations are using these sensors for highly accurate lighting and other appliances' control.

Ecoamica Technologies LLP shares a customer experience to explain. A Bengaluru-based technology multinational firm sought electricity consumption control solutions for areas like hosting server floors, corridors as well as aisles. The Ecosirius range of solutions by Ecoamica was installed throughout the intended premises with good results. A total of 300 sensors deployed throughout the area cut electricity usage from 12 hours down to just 1.5 hours a day. At an average electricity cost of ₹ 6.75 per unit, the firm's monthly electricity bill of ₹ 940,000 dramatically went down to ₹ 120,000—a

total saving of more than ₹ 800,000 in one month. The total setup cost was around ₹ 200,000, implying ROI within one month of the implementation.

The final word

When coupled with the right energy-efficient lighting and smart technology, PIR sensors can be a real money-saver. Sensor placements, proper usage and frequency of use play a big role in the ROI. Electricity can account for up to 20 per cent of an organisation's monthly expenses. So, controlling the consumption is a smart move to cut overhead costs. **EFY**

—Paromik Chakraborty, technical journalist, EFY

Retrofit LED Tubes For Energy-Efficient Lighting

➤ **LED tubelights consume about 60 per cent less power than their conventional counterparts**

If you still use incandescent or fluorescent tubelights in your office, it is time you switch to LED tubelights, which can save you up to 80 per cent power. Retrofit LED tubes, specifically Type-A ones, are the best choice for tubular lighting in any office, enterprise or industry as these are easy to install and have a life expectancy of 50,000 hours. Here we share some examples that prove the benefits of implementing LED lighting in work spaces.

Case 1. Pune-based Cummins India reported an ROI in 14 months and reduction of electricity bill by 50 per cent within two years of switching from traditional tubelights to easily-installable retrofit LED tubelights.

Kushal Uppal, founder and MD, IntelliZon, shares another example where a corporate establishment based in Hyderabad was looking to install solar panels on the rooftop to run their tubelights in the garage during daytime. Their initial plan was to install a 5kW solar system (including inverter) costing about ₹ 500,000 to run about a 100 tubelights. IntelliZon suggested the use of retrofit DC LED tubelights, which consumed about 60 per cent less power than conventional tubelights.

Elimination of a separate DC-AC converter for solar panels and tubelights and going straight with DC-DC (from solar panel to LED tubelight) enabled the company to install the entire system within ₹ 200,000, which was less than half of the initially planned capex. In the long run, there was a power saving of 30 per cent.

Case 2. US-based lighting solutions company Independence LED provided automobile giant BMW with LED solution kits, each comprising a pair of 15-watt, 2' retrofit LED tubes. The solution saved 34 watts per fixture, resulting in cumulative power savings of 53 per cent. By reducing the fixture wattage from 64 to 30, the annual savings were reported as 127.3kWh and \$11 per fixture. Over the 60,000-hour (16-year) lifespan of the LEDs, energy savings are over \$176 per fixture. Looking at all the data including power savings, reduced bulb replacement cost, lower maintenance expenses and labour cost savings, while also considering the average 3 per cent annual increase in the energy cost, the total savings is over \$348 per fixture. **EFY**

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